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From: Rhett Yardley, Dominic Hartman, Carter Novick, Justin Fox, Kaidan Holliday
Date: October 23rd, 2026
Subject: Project RAPTOR Progress Report

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1. Abstract

This midterm progress report details the initial development phase and technical milestones achieved for Project RAPTOR since the project proposal in September. Following project approval, the team focused on subsystem engineering, prototyping schedules, and funding procurement to support project viability.

Initial efforts centered on trade studies regarding materials, flight specifications, and cost optimization for both the launch vehicle and the deployable drone. The fixed-wing unmanned aerial vehicle (UAV) was modeled using Onshape CAD software, incorporating a pivoting wing mechanism designed to fit within the rocket's payload bay during ascent. This CAD geometry enabled preliminary volumetric, packaging, and aerodynamic assessments. Concurrently, the launch vehicle was modeled and simulated using OpenRocket trajectory software to verify aerodynamic stability, structural margins, and thrust requirements to achieve the target apogee.

Following design completion, prototype fabrication commenced. The drone's wing and swivel assembly was 3D printed using PETG filament to evaluate mechanism kinematics, deployment clearances, and structural durability, informing subsequent design revisions. In parallel, primary airframe components, including the nose cone, body tubes, and motor mounting assemblies have been procured, prioritizing cost reduction while maintaining structural integrity through fiberglass reinforcement.

Although the initial phase emphasized mechanical and aerodynamic design, preliminary research into avionics, power distribution, and flight-control integration is underway. Upcoming milestones include finalizing electrical systems, executing ejection tests, and conducting flight trials to validate autonomous drone stabilization, apogee deployment, and safe parachute recovery.

2. Objectives

The objective was to complete the initial designs for both drone and rocket, then begin obtaining necessary materials for construction. These technical objectives can be broken into a list of things that the team has worked towards since the project proposal:

- Designed a rocket in Open Rocket software and conducted simulations testing stability, strength, velocity, thrust, and acceleration using assorted styles and materials.
- Ordered finalized materials for rocket body; with construction starting soon.
- Developed RAPTOR Mini to allow for housing within the payload bay.
- 3D printed all drone segments for prototype testing and mechanism verification.
- Set up GoFundMe and contact multiple companies for financial support on the project.
- Contacted TRA Phoenix launch site to fully understand all requirements for both rocket and drone flights within their licensed airspace.

3. Introduction

This report for the RAPTOR (Rocket Apogee Payload Transition to Optimal Return) capstone serves as a formal progress report. Currently the overall goal, shown in figure 1, launches a rocket that deploys a drone at apogee. The governing body responsible for regulating this launch event would be Tripoli and Tripoli Phoenix, also known as TRAPHX [1]. TRAPHX is an organization that provides amateur rocket enthusiasts with a Phoenix launch site near Wickenburg and ensures that launches conducted there do not violate FAA (Federal Aviation Administration) and ATF (Alcohol, Tobacco, and Firearms) regulations. Tripoli achieves this through waivers such as the airspace waiver TRAPHX has which is 48,000' AGL (feet above ground level). [1]

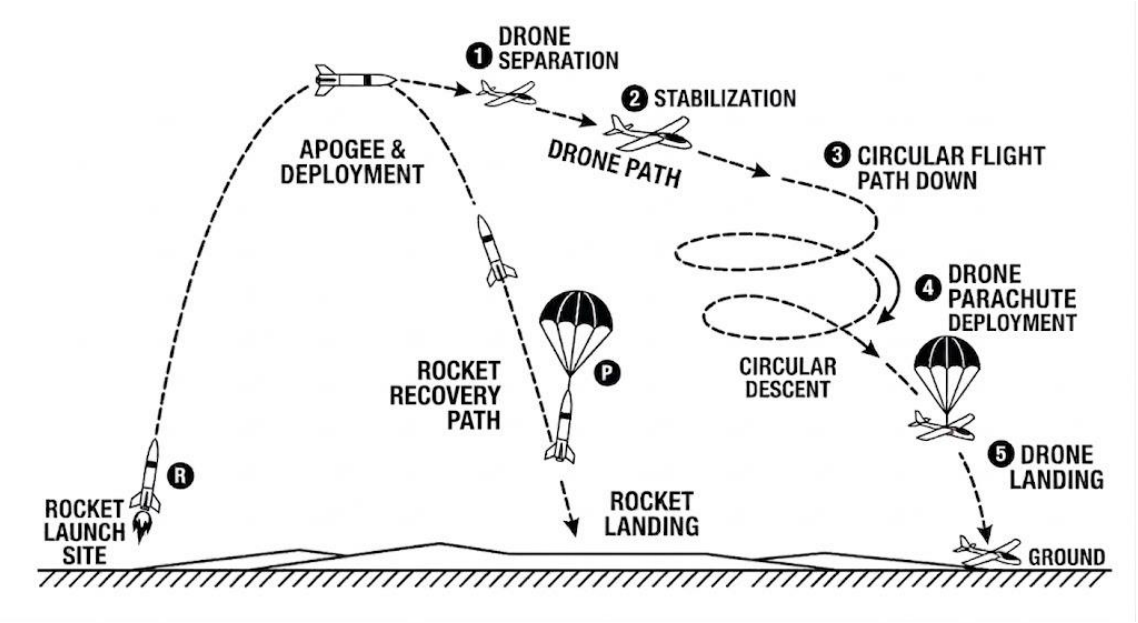


Figure 1: Overall goal for Project Raptor.

4. Tripoli Compliance

To ensure compliance with Tripoli and TRAPHX the following are needed.

- **Controlled Drone Deployment:** The Drone deployment needs to be manually triggered, during a launch a TRAPHX member will witness the rocket launch, when they give the okay, the pilot will need to deploy the drone. This will be done through extra servo channels on the flight controller. [1]
- **Launch Trajectories**: OpenRocket Simulator is a tool that lets you simulate rocket trajectories, center of gravity, and center of pressure for rocket designs. TRAPHX requires you to prove flight stability, launch speeds, and expected peak altitude. [1]This is most easily and historically done using OpenRocket Simulator through the NAU Rocket club.
- **Certifications**: To reach an apogee of 1,500 – 2,000 ft, a J or K class motor is most likely required. TRAPHX requires a level 2 certification for J class and above. [2]Team member Carter has this certificate due to his experience in the NAU rocket club.
- **FAA Part 107**: This is currently a work in progress; the FAA requires written test and drone registration for flying drones over 249g and those flying higher than 400 feet. FAA Part 107 certification will be completed prior to flight operations once testing funds are allocated.[1]

5. Tentative Schedule

October 2026

- Research
- Mid-term presentation
- Part ordering
- Initial designs
- Begin building prototypes

November 2026

- Initial Parts received
- Construction of main rocket systems
- Designing and refining designs
- Presentation of videos
- NAU Fest

December 2026

- Finalize first build.
- 1st Rocket Launch (without drone flight)
- Final Presentation

January 2027

- Revisions and testing of initial builds
- Version 2 designs
- Version 2 prototype

February 2027

- Final designs
- Final design construction
- Heavy Drone Testing

March 2027

- Components / systems refining.
- Mid-term presentations
- 2nd Rocket Launch

April 2027

- Final build and testing / tweaks to problems in last launch
- Final rocket launch

May 2027

- Final presentation

6. Development:

In accordance with the project schedule, the team has placed initial material orders for the rocket airframe structure (detailed in Section 7.2, Figure 3). To maximize structural strength while minimizing weight, the airframe will incorporate fiberglass reinforcement over rocket-grade kraft phenolic tubing, following proven manufacturing methods from the NAU Rocket Club. Project Advisor Dr. Carson Pete recommended utilizing pre-cut fiberglass fabric sheets wet-laid with epoxy resin and hardener as the preferred composite reinforcement method. Fundraising efforts included outreach to Novakinetics Aerosystems; while they could not sponsor the project directly due to existing capstone commitments, they provided technical consultation on composite tubing, fabric specifications, and design feedback.

Regarding launch vehicle avionics, the project sponsor provided all primary flight-control and tracking hardware. The avionics suite includes a Featherweight GPS Tracker and a Blue Raven flight computer. The programmable Blue Raven features an integrated inertial measurement unit (IMU), dual barometers, a high-G accelerometer, and four independent, software-configurable ordnance channels. These channels will execute sequential flight events, including apogee deployment and parachute recovery. Datasheet specifications indicate that current draw depends on battery voltage, with the board rated for up to 4.0 A total (~1.0 A per channel during simultaneous firing). While a 3.7 V LiPo battery is recommended, 9 V power supplies are also compatible.

For payload deployment, the sponsor recommended integrating a commercial-off-the-shelf RAPTOR CO_2 ejection system. This system utilizes electric matches (e-matches) to puncture pressurized CO_2 canisters, deploying the payload without the thermal hazards of black powder charges. Because standard e-matches typically require an all-fire current between 1.0 A and 2.0 A, further benchtop testing will be conducted to verify battery discharge capabilities and deployment reliability before launch.

7. Rocket Design:

Because housing and deploying a 30-inch-wingspan drone exceeds the capacity of standard high-powered rocket kits, the team engineered a custom launch vehicle using OpenRocket simulation software. The rocket is designed to carry the UAV to a target apogee of 1,500–2,000 ft AGL, eject the payload, and return safely under parachute recovery.

The airframe features a 7.51-inch (191mm) outer diameter divided into two primary modular sections:

1. Aft section: Houses the 54 mm motor mount tube (compatible with J- and K-class solid rocket motors), centering rings, recovery parachute, and dual deployment electronics bay.
2. Forward Section (Payload Bay): Houses the internal rail guide mechanism and stowed RAPTOR Mini UAV, sealed by the nose cone and actuated by the CO_2 ejection system.

7.1 Budgetary Constraints:

One of the main constraints on this project is cost. The heavier the rocket becomes, the more expensive the rocket motor is to propel this massive tube into the sky at 2000 ft AGL. The team's design relies on the rocket being as light and as inexpensive as possible. This includes making the drone out of a light-weight filament, making the rocket tube out of reinforced cardboard, using nylon bolts, etc.



Figure 2: Rocket Design

Figure 2 above shows the semi-final design for the rocket. The long tube towards the front section is the payload bay, the red section is the parachute and then electronics bay behind that, and finally rocket motor at the aft end. The diameter is 7.51” with a total length of 8.16 ft in length, when fully assembled. The weight for this design is around 25 lbs., with the team knowing this will not be an exact value.

7.2 Construction and Material Modifications:

The team has ordered parts for construction of the rocket shown in Figure 3; this includes the rocket nose cone and the tube section along with centering rings and an inner tube for the rocket motor mounting.

To cut costs and weight, the team will reinforce rocket grade cardboard with fiberglass. This is a common method used in high powered rocketry and has proven to be a durable yet semi-lightweight material. The team will also be constructing fins out of fiberglass over wood, a little heavier and stronger material for the high forces acting on the fins. The team also sourced a Blue Raven and a Featherweight Tracker for the rocket flight electronics that reduces the $400+ cost that the team would otherwise need to take of the limited budget.

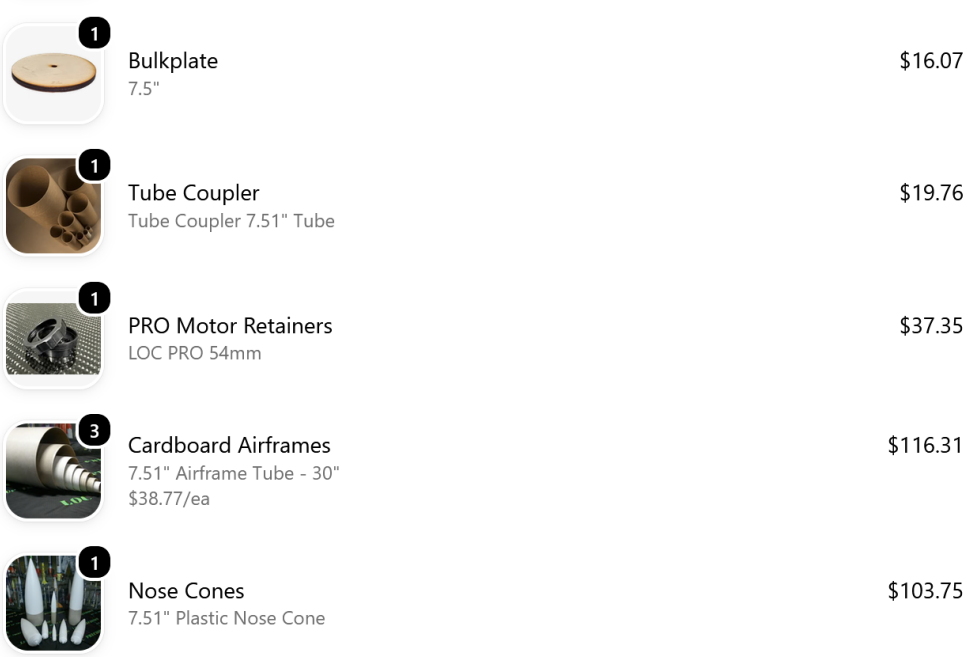


Figure 3: Rocket Components Order

7.3 Stability Concerns and Resolutions:

With everything mentioned, the stability ratio shown in the top right of figure 2 is the most important aspect. Every time something changes in weight or in position within the rocket, the center of gravity moves. (labeled as the blue dot in figure 2) The center of pressure is right behind that (red dot) and is heavily reliant on the fins and rocket motor. This stability number needs to be between 1 and 2 calibers, meaning that the center of gravity must be a certain percentage forward of the center of pressure. This creates a balance point that is the pivot point for any movement in the rocket. To complete the flight these aspects of the rocket will be verified on site with the range safety officer (RSO) to ensure that the rocket is safe to fly and will not suddenly turn midair.

The rocket design has taken in every aspect of weight, pressure, strength and durability to ensure that the drone can get to apogee successfully.

7.4 Rocket Simulations:

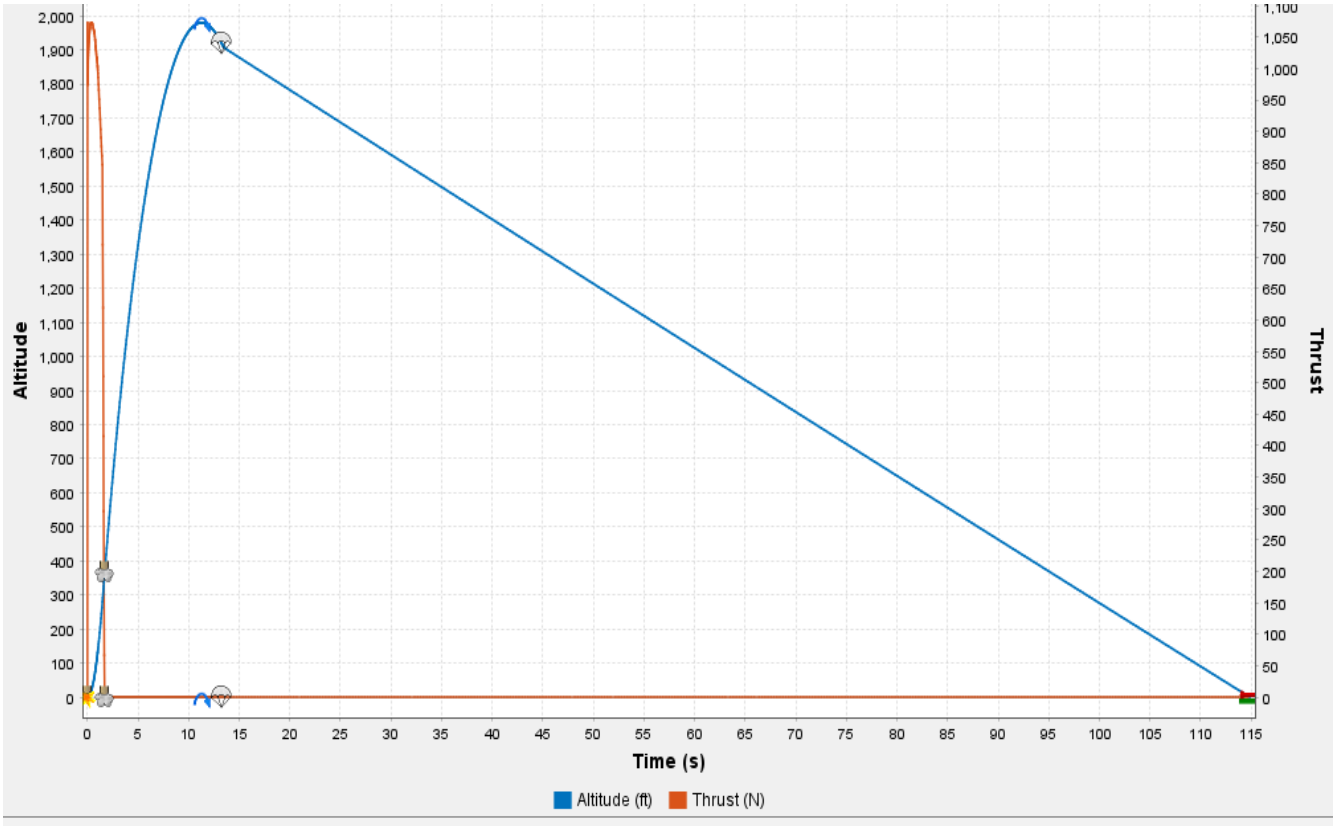


Figure 2: Rocket Altitude & Thrust vs Time

In figure 4 shown above, the rocket is getting to about 2000 ft AGL after about 12 seconds while also showing the rocket motor thrust and motor cutout.

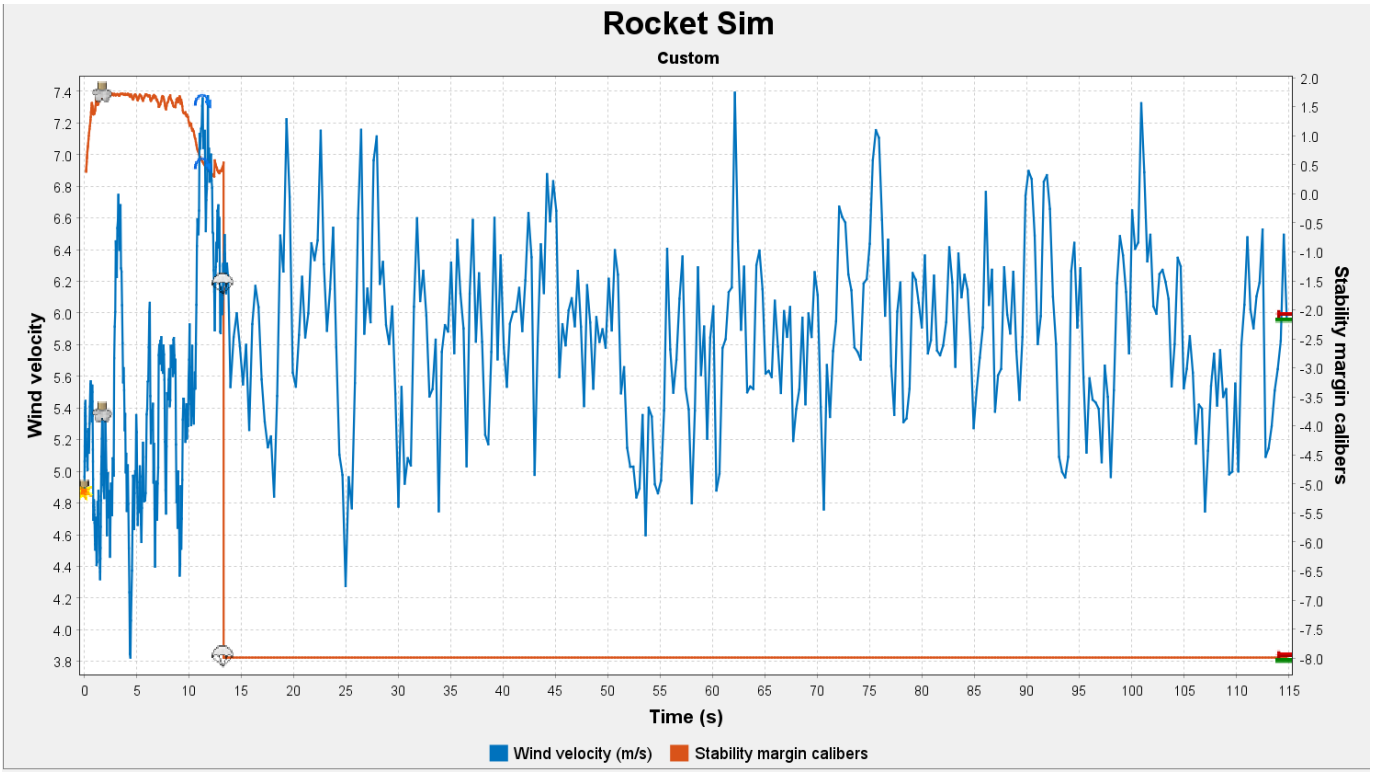


Figure 3: Variable Wind Speed at Altitude and Stability Margins

This simulation from figure 5 shows stability through a variable wind speed. This shows the rocket staying stable at 13 mph, 90-degree wind that increases with altitude after launch. This shows rocket flight stability through tougher conditions and clears up concerns the team had about restrictive launch conditions that would otherwise ground the rocket during the windier months at the end of the Spring semester.

8. RAPTOR Mini Subsystem Design

8.1 Subsystem Overview

RAPTOR Mini serves as the deployable fixed wing drone being developed as part of Project RAPTOR. The unmanned aerial vehicle (UAV) is designed to be housed within the rocket during launch, deployed at apogee, and configured for controlled flight after separation. The drone must fit within the available payload bay dimensions, withstand the mechanical demands associated with deployment, and sustain controlled flight after separation from the rocket. These requirements influence the airframe geometry, structural reinforcement, material selection, and integration of electrical and electronic systems.

Initial development began by translating customer requests into engineering constraints. These requirements dictated the size, shape, and overall design of the airframe. The team carried out the next stage of design utilizing computer-aided design (CAD) software to model the drone. Using CAD allowed for the UAV's geometry, component placement, and folding mechanisms to be evaluated. Decision matrices were then used for comparison of the airframe configurations, material selection, and flight controllers to guide the next stages of development.

Three-dimensional (3D) printing was selected as the primary manufacturing method to support rapid prototyping and iterative design. The manufacturing method heavily influenced the UAV's design as the build plate size restricted the maximum component length to 220 mm. Segmented modeling was employed to combat this impediment, which required reinforced connection points. Once the team achieved sufficient structural design while meeting packaging and deployment requirements, RAPTOR Mini Version 1 was printed using polyethylene terephthalate glycol (PETG). The prototyping stage of design paved the way for assembly and evaluation. This prototype provides a physical reference for identifying areas that require refinement before manufacturing the next iteration.

The next design iteration is planned to use ASA Aero filament and incorporate carbon-fiber tubes for enhanced structural reinforcement. Control surfaces such as ailerons, and mounting provisions for the required electronics must be added prior to flight. Furthermore, the remaining work includes refining the airframe, integrating the flight-control system, and conducting flight testing to evaluate the drone's structural integrity, deployment, and flight performance. This report documents the current design, engineering decisions, prototype development, and remaining work required to advance RAPTOR Mini towards a flight-ready configuration.

8.2 Engineering Constraints

RAPTOR Mini's design is driven by its integration with the rocket and the requirements of the intended flight environment. The primary engineering considerations are packaging, structural integrity, material selection, manufacturing, and flight-control integration.

8.2.1 Packaging and Deployment

The drone must fit within the available internal volume of the rocket and remain compatible with the deployment mechanism. The airframe geometry, component placement, and assembly interfaces must accommodate the stowed configuration without interfering with deployment.

The design must also allow the drone's components to remain securely positioned during launch while permitting the mechanism to deploy as intended. These requirements influence the airframe layout, component mounting locations, and interfaces between the drone and the deployment system.

8.2.2 Structural Design

The airframe must maintain sufficient stiffness and structural integrity during handling, deployment, and flight. Carbon fiber tubes are integrated into the design to reinforce the structure while limiting additional mass. Their placement and attachment points must provide adequate support without adding additional complexities. The printed components must also provide suitable mounting locations for the wings, control surfaces, and electronic hardware.

8.2.3 Material Selection and Manufacturing

PETG was used to manufacture the initial prototype for assembly evaluation. ASA Aero has been selected for the next iteration. The revised design will be evaluated for print quality, dimensional accuracy, component fit, and structural suitability.
The transition to the next material will provide an opportunity to refine component geometry and address any issues identified during initial assembly. Manufacturing considerations include print orientation, component dimensions, assembly interfaces, and the ability to incorporate structural reinforcement.

8.2.4 Flight Control and Electrical Integration

The flight-ready design will incorporate ailerons and the associated actuation hardware, along with electronics required for flight control and propulsion. The airframe must provide sufficient space for component mounting and wiring while maintaining structural integrity and compatibility with the deployment system.

Electrical integration will require finalizing component selection, defining mounting locations, and establishing power and signal connections. Control surface movement and electrical operation will be verified during ground testing before the system progresses to flight testing.

8.3 Design Methods

Following the selection of high-level engineering constraints to guide the drone's design described in Section 2, the team began translating the customer requirements and engineering constraints into a practical design approach. Initial design discussion focused on how to develop an airframe that could meet the project's packaging, structural, and deployment requirements while remaining feasible to manufacture and evaluate within the project timeline.

The team first considered how the drone would fit within the rocket. The arrangement of the wing mounting mechanism and electronic component placement within the stowed and deployed configurations was the next blocker. These considerations guided the selection of the airframe architecture and the initial geometry development. The team used computer-aided design (CAD) software to model the airframe and methodically review component placement, overall dimensions, and assembly interfaces before manufacturing.

The drone team selected 3D printing as the primary manufacturing method for the initial prototype. This approach allows the drone to be modeled in software, physically manufactured, then assessed. This significantly improves the team's ability to review geometry, component placement, and assembly connections through prototyping. The design iterations can be printed, assembled, evaluated, and revised as needed throughout development. Rapid manufacturing was also identified as an important consideration because components may need to be replaced if damaged during flight or testing. Using 3D printing allows revised or replacement parts to be manufactured without prolonged down time, supporting iterative design and reducing delays between development stages. This approach was selected to help the team refine the drone's mechanical design while maintaining progress towards the launch windows.

The initial CAD design was manufactured using PETG to evaluate component fit and the overall assembly. Information obtained during prototype assembly will guide modifications to the next design iteration, which is planned to use ASA Aero and incorporate the additional structural and flight-control components required for the flight ready configuration.

8.4 Decision Matrices

After establishing the initial design approach, the team identified three critical design decisions that would benefit from a structured comparison of alternatives. Decision matrices were developed to evaluate these decisions against engineering criteria derived from the design requirements. The matrices provide a consistent method for comparing alternatives; accounting for the relative importance of each criterion while documenting the decision-making process. The three decision matrices address airframe architecture, airframe material, and the Flight Controller.

Each decision matrix uses a weighted scoring method to compare the available alternatives. Evaluation criteria are assigned weights based on their relative importance, while each alternative receives a rating for each criterion. The rating for each criterion is multiplied by its weight, and the resulting values are summed to calculate the alternative's total weighted score. Seen here below:

$$Total\ Score = \sum Rating \times Weight$$

The ratings are assigned on a scale of one to five, with the interpretation of each rating defined by the team's evaluation method. Weights represent the relative importance of the criteria and should sum to 100% when expressed as percentages. The alternative with the highest total weighted score is selected as the preferred option for further development.

The criteria selected for each matrix reflect the engineering requirements and constraints established during the initial design stage. For airframe architecture, the team considered packaging, structural performance, mass, assembly, and integration. Material selection focused on manufacturing quality, structural suitability, cost, and ease of fabrication. Flight controller selection considered physical size, power distribution, firmware compatibility, sensor support, telemetry capabilities, and reliability. These criteria allowed the team to compare each alternative based on its suitability for RAPTOR Mini's intended operation.

The results of the decision matrices were used to guide the team's design selections and establish a direction for continued development. Each selected alternative represents the option with the highest weighted score based on the criteria and ratings assigned by the team. These selections provided a foundation for developing the CAD model and manufacturing the initial prototype. However, the matrices represent preliminary evaluations and do not independently verify the performance of the selected components or configurations. Further evaluation through prototype assembly, ground testing, and subsequent flight testing will be necessary to determine whether the selected designs satisfy the project's engineering requirements.

8.4.1 Airframe Architecture Selection

Four preliminary airframe configuration concepts were evaluated to determine the optimal design. Concept A is a Solid Rotating High Wing, where the wing pivots at a center point. Concept B is a drone integrated into the rocket frame, where the wings do not fold. Concept C is a scissor folding wing mechanism. Concept D is a telescopic wing mechanism. Candidate configurations will be evaluated based on packaging, structural performance, mass, assembly, and integration requirements. The alternatives were evaluated against criteria associated with packaging, structural performance, mass, assembly, and integration. These criteria reflect the need to fit the drone within the rocket while maintaining a practical mechanism for deployment and subsequent flight. Figure 6 represents the methodical approach behind the wing configuration decision process on RAPTOR Mini.

RAPTOR Mini Wing Structure Decision Matrix

CRITERIA	WEIGHTAGE	Concept A: Solid Rotating High Wing		Concept B: Fixed-Wing Rocket Integrated Frame		Concept C: Scissor Folding Wings		Concept D: Telescopic Collapsing Wings	
		RATING	TOTAL	RATING	TOTAL	RATING	TOTAL	RATING	TOTAL
Structural Integrity	12%	5	12.00%	4	9.60%	2	4.80%	2	4.80%
Structural Rigidity	12%	5	12.00%	4	9.60%	2	4.80%	2	4.80%
Deployment Reliability	12%	5	12.00%	5	12.00%	3	7.20%	1	2.40%
Deployment Simplicity	12%	5	12.00%	5	12.00%	2	4.80%	1	2.40%
Mass Efficiency	10%	4	8.00%	2	4.00%	3	6.00%	2	4.00%
Volumetric Efficiency	10%	3	6.00%	2	4.00%	4	8.00%	4	8.00%
Payload Capacity	8%	5	8.00%	3	4.80%	4	6.40%	3	4.80%
Aero Performance	8%	5	8.00%	3	4.80%	2	3.20%	3	4.80%
Ease of Assembly	8%	4	6.40%	1	1.60%	2	3.20%	2	3.20%
Maintenance Capability	8%	4	6.40%	1	1.60%	2	3.20%	2	3.20%
Total	100%		TOTAL		TOTAL		TOTAL		TOTAL
			90.80%		64.00%		51.60%		42.40%

Figure 4: Decision Matrix 1 for the main wing design.

Based on Figure 6 the Solid Rotating High Wing configuration received the highest overall score and was selected for further development. This selection established the preferred airframe configuration for the design stage. The configuration must still be evaluated for packaging, reliable deployment, structural stability, and compatibility with the final flight control system.

8.4.2 Airframe Material Selection

The material selection process considers structural requirements, environmental conditions, manufacturing quality, and mass. PETG was used as the initial prototyping material due to ease of printing, affordability, and availability. This material provided a practical means of evaluating the design and assembly before manufacturing the next iteration. The four materials considered were ASA Aero, ASA-GF, PLA Aero, and Standard PETG.

RAPTOR Mini Filament Material Decision Matrix

CRITERIA	WEIGHTAGE	Concept A: ASA Aero RATING	Concept A: ASA Aero TOTAL	Concept B: ASA-GF RATING	Concept B: ASA-GF TOTAL	Concept C: PLA Aero RATING	Concept C: PLA Aero TOTAL	Concept D: Standard PETG RATING	Concept D: Standard PETG TOTAL
Volumetric Efficiency	25%	5	25.00%	2	10.00%	5	25.00%	1	5.00%
Structural Strength	20%	4	16.00%	5	20.00%	2	8.00%	3	12.00%
UV Resistance	15%	5	15.00%	5	15.00%	2	6.00%	3	9.00%
Heat Deflection	15%	5	15.00%	5	15.00%	1	3.00%	3	9.00%
Structural Rigidity	10%	3	6.00%	5	10.00%	3	6.00%	3	6.00%
Layer Adhesion	10%	4	8.00%	4	8.00%	3	6.00%	5	10.00%
Printability	5%	2	2.00%	1	1.00%	3	3.00%	5	5.00%
Total	100%		TOTAL		TOTAL		TOTAL		TOTAL
			87.00%		79.00%		57.00%		56.00%

Figure 5: Decision Matrix 2 for the material selection.

Figure 7 represents the methodical approach to determining which 3D printable material will be used for the final design manufacturing of RAPTOR Mini. The final iteration will provide an opportunity to evaluate the selected material in the revised airframe geometry and determine whether the printed components meet the design requirements. The material selection will be further assessed through manufacturing, assembly, and subsequent testing.

8.4.3 Flight Controller Selection

The flight controller decision matrix evaluated the alternatives based on power distribution, physical size, firmware compatibility, sensor support, telemetry capabilities, and reliability. These criteria were selected to assess each controller's suitability for the available airframe space, as well as electrical and control requirements for RAPTOR Mini. The alternatives included the Matek F405-WING V2, a SpeedyBee F405-based wing controller, a PiHawk 6C Mini, and a SpeedyBee F4 Mini.

RAPTOR Mini Flight Computer Decision Matrix

CRITERIA	WEIGHTAGE	Concept A: Matek F405 Wing V2		Concept B: SpeedyBee F405 WING APP		Concept C: Pihawk 6C Mini		Concept D: SpeedyBee F4 Mini	
		RATING	TOTAL	RATING	TOTAL	RATING	TOTAL	RATING	TOTAL
Power Distribution	25%	5	25.00%	5	25.00%	4	20.00%	1	5.00%
Physical Dimensions	20%	5	20.00%	4	16.00%	2	8.00%	4	16.00%
Firmware Support	20%	5	20.00%	5	20.00%	5	20.00%	3	12.00%
Onboard Sensors	15%	4	12.00%	4	12.00%	5	15.00%	3	9.00%
Telemetry Support	10%	5	10.00%	5	10.00%	5	10.00%	2	4.00%
Product Reliability	10%	5	10.00%	4	8.00%	5	10.00%	3	6.00%
Total	100%		TOTAL		TOTAL		TOTAL		TOTAL
			97.00%		91.00%		83.00%		52.00%

Figure 6: Decision matrix 3 for the flight computer.

Figure 8 depicts the decision-making process behind determining that the Matek F405-WING V2 is the most ideal candidate. It received the highest overall score therefore it was selected as the preferred flight controller. This decision establishes the intended controller for the next stage of development. Final integration will require confirming firmware compatibility, available inputs and outputs, power requirements, sensor support, and compatibility with the selected propulsion and control-surface hardware.

8.5 Prototype Fabrication

The prototype was fabricated out of PETG to evaluate assembly and kinematics as the primary objective. Minor issues presented with geometry and rotation mechanism as initial components did not function correctly. Iterations in the wing and swivel design were made to correct this. In relation to 3D printing, the design requires modifications to be functionally printed in ASA AERO as the filament does not function well with support structures. However, the weight savings using the ASA AERO was greater than expected with approximately 60% reduction in weight. While critical stress bearing components may still need to be built with an auxiliary material to manage the action of deploying the wings, the weight of the design is now sufficient to sustain stable flight.



Figure 7: Prototype PETG Drone, Open



Figure 8: Prototype PETG Drone, Closed

8.6 Results

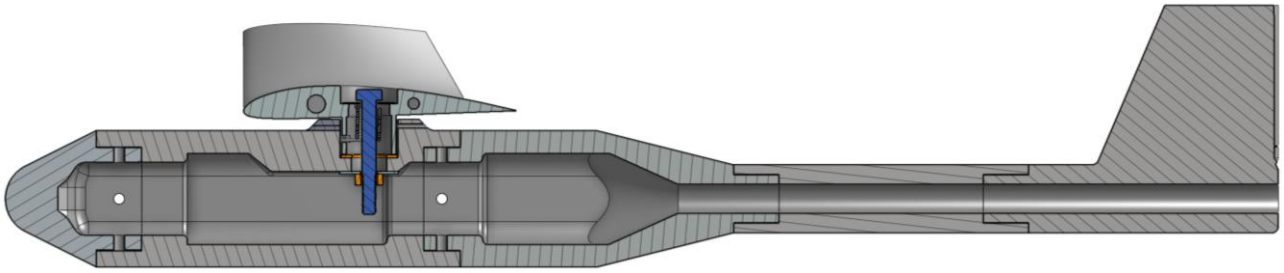


Figure 9. Cross sectional view of the Raptor Mini.

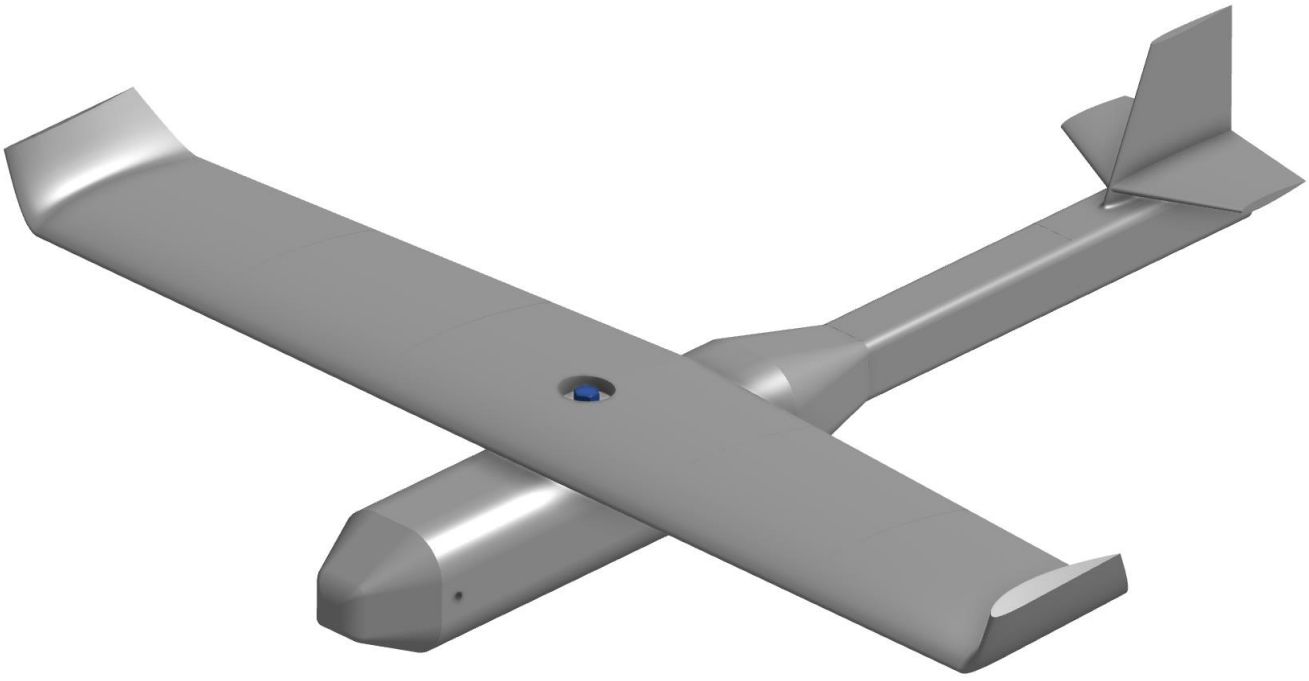


Figure 10: 3D view of the Raptor Mini.

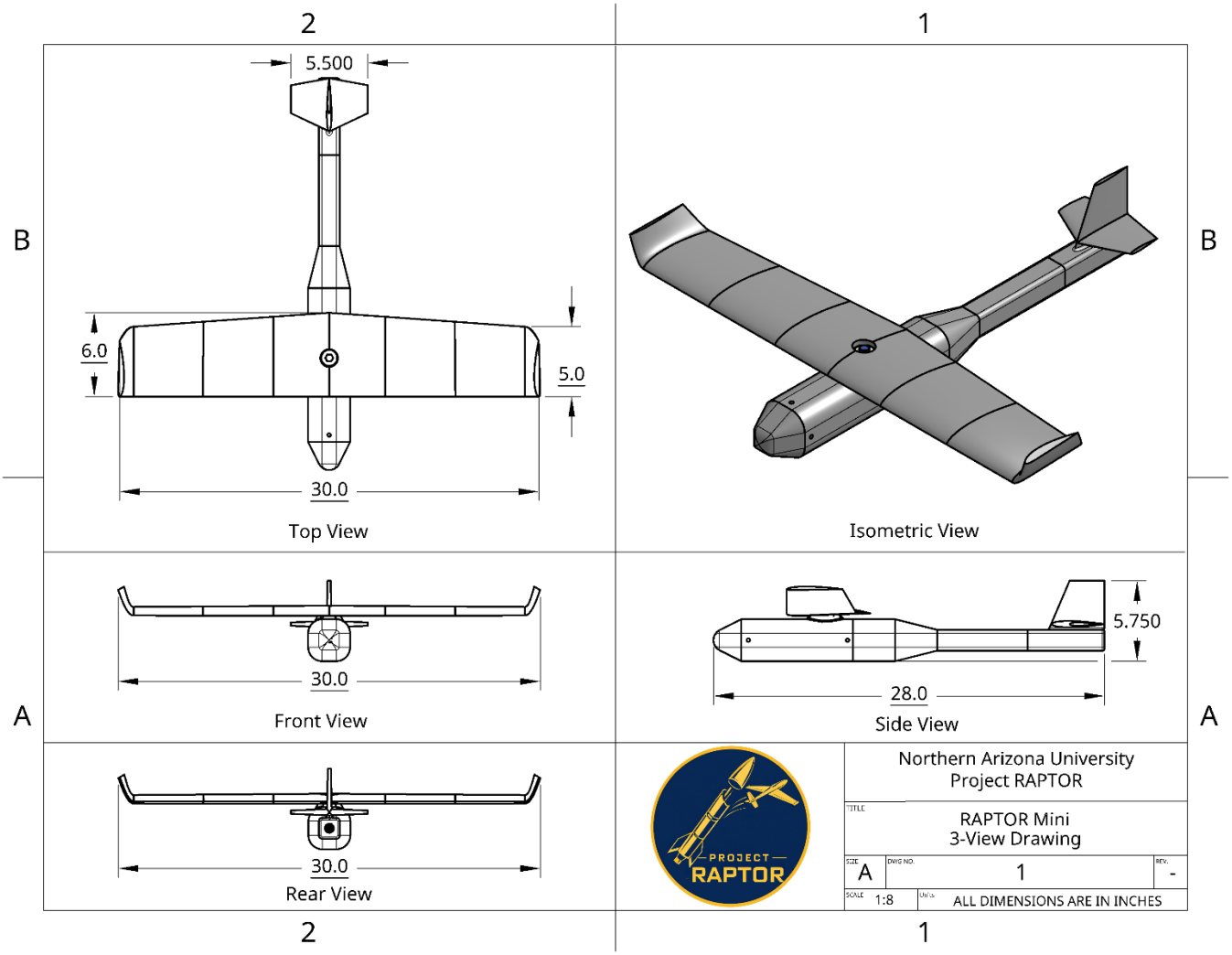


Figure 11: Engineering drawing of the Raptor mini with the wings unfolded.

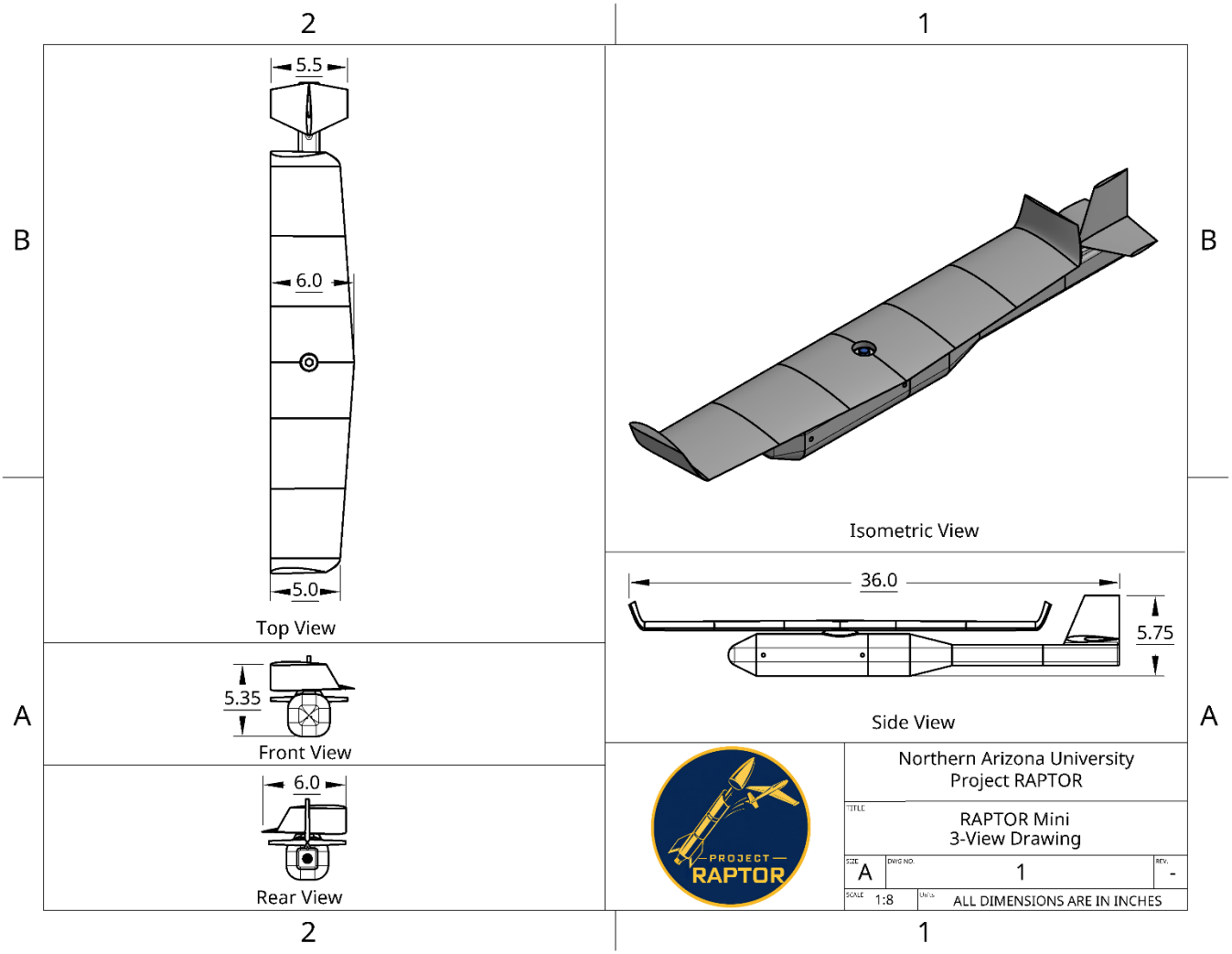


Figure 12: Engineering drawing of the Raptor mini with the wings folded.

8.7 Current Bill of Materials:

PARTS & SUPPLY REQUEST

Please provide a detailed description for each request item and include weblink whenever possible.

Vendor Name	Weblink to Item	Description	Item or Catalog #	Size/ Color	Qty	Discount Code	Total Cost (including tax & shipping)
TinderRocketry	**nderrocketry.com/the-rapto**	CO2 Ejection System	RAPTOR	Flanged	1		**$ 178.50**
Bambu Labs	**tore.bambulab.com/produc**	ASA Aero Filament	46100	White	1		$ 49.99
Creality	rww.amazon.com/dp/B0FS	Creality PETG 2kg	Grey/Grey	rey/Gre	1		$ 28.49

Figure 13: NAU Order Form

Drone/ Glider		
Subsystem	Component	Link
Flight Controller	Mateksys F405-WMN or F405-WING-V2 or Slim	https://www.amazon.com/dp/B0GRT4GZM8
ESC	Holybro Tekko32 F4 35A	https://holybro.com/products/tekko32-f4-45a-esc
Motor	SunnySky X2207S 2100KV Brushless Motor CW	https://www.buddyrc.com/products/sunnysky-x2207s-2100kv-brushless-motor-ccw-on-sale
Propeller System	Folding Pusher Prop	https://icare-icarus.3dcartstores.com/aero-naut-Cam-Carbon-blades-6-x-3_p_2997.html
Actuators	EMAX ES08MA II (x2 to x4)	https://www.amazon.com/dp/B0DN1G3P2M
Battery	Main Flight Pack	https://a.co/d/00ir4yKE
Power Smoothing	Low-ESR Capacitor	https://a.co/d/08G3sfO5
GPS / Magnetometer	HGLRC M100-5883 GPS	https://www.hglrc.com/products/m100-5883-gps?srsltid=AU7gw4VrG17M36aq5GEHMwZolW
RC Receiver	Radiomaster RP1 or EP1	https://pyrodrone.com/products/radiomaster-rp1-elrs-2-4ghz-nano-receiver-w-ufl-antenna
FPV Camera	Predator 5 Mini 1.8mm from Foxeer	we have this already
Video Transmitter	Foxeer Reaper Nano V2 VTx 5.8G	https://www.foxeer.com/foxeer-reaper-nano-v2-vtx-5-8g-72ch-350mw-tramp-g-583
VTX Antenna	FOXEER Lollipop FPV U.FL Antenna 5.8G	https://a.co/d/01KPfx88
Data Logger	featherwight tracker	we have this already
Logger Battery	single-cell 1S LiPo	https://a.co/d/09vRX3Q9
Beeper	want a beeper for finding it easier?	
Rocket		
Subsystem	Component	Link
Fuselage	Cardboard 7.51" Airframe Tube - 30" x3	https://locprecision.com/products/cardboard-airframes?variant=39778656256191
Coupler	Tube Coupler 7.51" Tube	https://locprecision.com/products/tube-coupler?variant=39778656354495
Bulkplate	Bulkplate 7.5"	https://locprecision.com/products/bulkplate?variant=40823480877247
Centering Rings	CR-7.51-54mm Centering Rings	https://locprecision.com/products/751rings?variant=39778660221119
Nose Cone	7.51" Plastic Nose Cone	https://locprecision.com/products/nose-cones?variant=39778657173695
Motor Retainer	LOC PRO 54mm Motor Retainer	https://locprecision.com/products/loc-pro-motor-retainer?variant=39778649407679
Motor Tube	FT22-STD-180-NAT Fiberglass Motor Tube	https://www.madcowrocketry.com/2-2-54mm-g12-motor-tube/
CO2 Ejection System	RAPTOR CO2 Ejection System Flanged	https://www.tinderrocketry.com/the-raptor-co2-ejection

Figure 14: Working BOM

9. Conclusion:

With steady progress and valuable lessons learned from prototyping, Project RAPTOR remains on schedule with adequate contingency for revisions. The next steps of the project will be taking the recently ordered components to then modify and assemble the rocket with plans on attempting a launch in December or even earlier if timing permits. Once sections of the rocket are completed, further testing can be conducted to include the drone deployment, parachute ejection, and data collection from onboard telemetry to determine if any design or operation modifications are required.

In the aspect of Drone deployment, the drone will be launched from the Payload Bay section of the rocket using the CO2 ejection system. The team will perform a ground test of the payload deployment mechanism to ensure reduced losses in the case of any catastrophic damage. These CO2 ejection tests will ensure effective ejection of the nosecone as well as sufficient clearance with the deployment of the drone. Additionally, there will be testing on the parachute ejection in the aft section of the rocket. This will include a similarly weighted front section to emulate a fully loaded rocket and ensure that even during a “No-Go" drone deployment event, the rocket can effectively and safely deploy the parachute to recover the entire system with minimal damage to the system. The main milestones are finalizing construction and certifying the rocket for a safe flight before the December launch window.

Lastly the drone itself will go through several iterations as control and communications systems are integrated as well as determining how many structural components will be printed in the ASA AERO material compared to PETG or similar for structural strength. Test glider flights and controlled flights of the drone are the next milestone for that component of the RAPTOR project alongside event testing of the rocket system.

Overall, the RAPTOR project is running smoothly, and the team is looking forward to reaching the next milestone for each component of the project.

References

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